Version control systems

- Git
- GitHub – Cloud-based Git repository hosting
- GitLab – Git repository with built-in CI/CD pipelines
- Bitbucket – Git repository with Jira integration

RATNADEEP SALI

Ratnadeep sali

- **Git**

1. Git Setup and Configuration

git --version # Check Git version

git config --global user.name "Your Name" # Set global username

git config --global user.email "your.email@example.com" # Set global email git config --global core.editor "vim" # Set default editor

git config --global init.defaultBranch main # Set default branch name

git config --list # View Git configuration

git help <command> # Get help for a Git command

---

2. Creating and Cloning Repositories

git init # Initialize a new Git repository

git clone <repo_url> # Clone an existing repository

git remote add origin <repo_url># Link local repo to a remote repo

git remote -v # List remote repositories

---

3. Staging and Committing Changes

git status # Check the status of changes

git add <file> # Add a file to the staging area git add . # Add all files to the staging area

git commit -m "Commit message" # Commit staged files

git commit -am "Commit message" # Add & commit changes in one step git commit --amend -m "New message" # Modify the last commit message

RATNADEEP SALI

4. Viewing History and Logs

git log # Show commit history

git log --oneline # Show history in one-line format

git log --graph --decorate --all # Display commit history as a graph git show <commit_id> # Show details of a specific commit git diff # Show unstaged changes

git diff --staged # Show staged but uncommitted changes git blame <file> # Show who modified each line of a file

5. Branching and Merging

git branch

List all branches

git branch <branch_name>

Create a new branch

git checkout <branch_name>

Switch to another branch

git checkout -b <branch_name>

Create and switch to a new branch

git merge <branch_name>

Merge a branch into the current branch

git branch -d <branch_name>

Ratnadeep sali

Delete a local branch

git branch -D <branch_name>

Force delete a branch

---

6. Working with Remote Repositories

git fetch # Fetch changes from remote repo

git pull origin <branch_name> # Pull latest changes

git push origin <branch_name> # Push changes to remote repo git push -u origin <branch_name> # Push and set upstream tracking git remote show origin # Show details of the remote repo git remote rm origin # Remove the remote repository

---

7. Undoing Changes

git restore <file> # Unstage changes

git restore --staged <file> # Unstage a file from the staging area

reset HEAD~1 # Undo the last commit but keep changes

git reset --hard HEAD~1 # Undo the last commit and discard changes git revert <commit_id> # Create a new commit to undo changes

git stash # Save uncommitted changes temporarily

git stash pop # Apply stashed changes

git stash drop # Remove stashed changes

---

8. Tagging and Releases

git tag # List all tags.

git tag <tag_name> # Create a tag .

git tag -a <tag_name> -m "Tag message" # Create an annotated tag.

git push origin <tag_name> # Push a tag to remote.

git tag -d <tag_name> # Delete a local tag .

git push --delete origin <tag_name> # Delete a remote tag.

---

9. Working with Submodules

git submodule add <repo_url> <path> # Add a submodule

git submodule update --init --recursive # Initialize and update submodules git submodule foreach git pull origin main # Update all submodules

---

10. Git Aliases (Shortcuts)

git config --global alias.st status # Create alias for status command

git config --global alias.co checkout # Create alias for checkout command git config --global alias.br branch # Create alias for branch command git config --global alias.cm commit # Create alias for commit command git config --list | grep alias # View all configured aliases

---

11. Deleting Files and Folders

git rm <file> # Remove a file and stage deletion

git rm -r <folder> # Remove a directory

Ratnadeep sali

```
git rm --cached <file>          # Remove file from repo but keep locally
```

---

12. Force Push and Rollback (Use with Caution)

```
git push --force                # Force push changes

git reset --hard <commit_id>    # Reset repo to a specific commit   git reflog
# Show history of HEAD changes

git reset --hard ORIG_HEAD      # Undo the last reset
```

Ratnadeep sali

2. GitHub

Authentication & Configuration

gh auth logout – Log out of GitHub CLI

gh auth status – Check authentication status

gh auth refresh – Refresh authentication token

gh config set editor <editor> – Set the default editor (e.g., nano, vim) gh config get editor – Get the current editor setting

Repository Management

gh repo list – List repositories for the authenticated user

gh repo delete <name> – Delete a repository

gh repo rename <new-name> – Rename a repository

gh repo fork --clone=false <url> – Fork a repository without cloning

Branch & Commit Management

gh branch list – List branches in a repository

gh branch delete <branch> – Delete a branch gh browse – Open the repository in a browser gh co <branch> – Check out a branch

Issue & Pull Request Handling

gh issue create – Create a new issue

gh issue close <issue-number> – Close an issue

gh issue reopen <issue-number> – Reopen a closed issue

gh issue comment <issue-number> --body "Comment" – Add a comment to an issue

gh pr list – List open pull requests

gh pr checkout <pr-number> – Check out a pull request branch

gh pr close <pr-number> – Close a pull request

Ratnadeep sali

Gists & Actions

gh gist create <file> – Create a new gist

gh gist list – List all gists

gh workflow list – List GitHub Actions workflows

gh run list – List workflow runs

Webhooks:

- Go to **Repo → Settings → Webhooks → Add Webhook**
- Events: push, pull_request, issues, etc.
- Payload sent in JSON format to the specified URL

---

3. GitLab

Commands

Project & Repository Management

gitlab project create <name> – Create a new project gitlab
repo list – List repositories

gitlab project delete <id> – Delete a project

gitlab repo fork <repo> – Fork a repository

gitlab repo clone <url> – Clone a GitLab repository gitlab
repo archive <repo> – Archive a repository

Issues & Merge Requests

gitlab issue list – List all issues in a project

gitlab issue create --title "<title>" --description "<desc>" – Create an issue gitlab issue close
<issue_id> – Close an issue

gitlab issue reopen <issue_id> – Reopen an issue

gitlab merge_request list – List merge requests

gitlab merge_request create --source-branch <branch> --target-branch <branch>

--title "<title>" – Create a merge request

gitlab merge_request close <mr_id> – Close a merge request

Pipeline & CI/CD

gitlab pipeline trigger – Trigger a pipeline

gitlab pipeline list – List all pipelines

gitlab pipeline retry <pipeline_id> – Retry a failed pipeline gitlab pipeline cancel <pipeline_id> – Cancel a running pipeline gitlab pipeline delete <pipeline_id> – Delete a pipeline

gitlab runner register – Register a CI/CD runner

gitlab runner list – List registered runners

gitlab runner unregister <runner_id> – Unregister a runner

User & Group Management

gitlab user list – List users in GitLab

gitlab user create --name "<name>" --email "<email>" – Create a new user gitlab group list – List groups in GitLab

gitlab group create --name "<group_name>" --path "<group_path>" – Create a group

gitlab group delete <group_id> – Delete a group

Access & Permissions

gitlab project member list <project_id> – List project members

gitlab project member add <project_id> <user_id> <access_level> – Add a user to a project

gitlab group member list <group_id> – List group members

gitlab group member add <group_id> <user_id> <access_level> – Add a user to a group

Repository Protection & Settings

gitlab branch protect <branch> – Protect a branch

gitlab branch unprotect <branch> – Unprotect a branch gitlab repository mirror – Set up repository mirroring

Ratnadeep sali

gitlab repository settings update – Update repository settings

Webhooks:

- Go to **Settings → Webhooks**
- Select triggers: Push events, Tag push, Merge request, etc.
- Use GitLab CI/CD with .gitlab-ci.yml

---

4. Bitbucket

Commands

Repository Management

- bitbucket repo create <name> – Create a repository ● bitbucket repo list – List all repositories
- bitbucket repo delete <name> – Delete a repository
- bitbucket repo clone <repo-url> – Clone a repository
- bitbucket repo fork <repo> – Fork a repository
- bitbucket repo update <repo> – Update repository settings

Branch Management

- bitbucket branch create <branch-name> – Create a new branch
- bitbucket branch list – List all branches
- bitbucket branch delete <branch-name> – Delete a branch

Pipeline Management

- bitbucket pipeline run – Run a pipeline
- bitbucket pipeline list – List pipelines
- bitbucket pipeline stop <pipeline-id> – Stop a running pipeline
- bitbucket pipeline rerun <pipeline-id> – Rerun a pipeline

Issue Tracking

- bitbucket issue list – List all issues
- bitbucket issue create "<title>" --kind=<bug/task/enhancement> – Create an issue
- bitbucket issue update <issue-id> --status=<open/closed/resolved> – Update issue status
- bitbucket issue delete <issue-id> – Delete an issue

Pull Request Management

- bitbucket pullrequest create --source <branch> --destination <branch> – Create a pull request
- bitbucket pullrequest list – List pull requests
- bitbucket pullrequest merge <id> – Merge a pull request
- bitbucket pullrequest approve <id> – Approve a pull request
- bitbucket pullrequest decline <id> – Decline a pull request **Webhooks:**
    - Go to **Repo → Repository Settings → Webhooks**
        - Choose event types like repo:push, pullrequest:created, etc.